



Dissecting the Dynamics of EGFR Down Modulation Following osimertinib Exposure in Lung Adenocarcinoma

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1. Introduction

1.1. Lung Adenocarcinoma

Lung cancer is one of the leading causes of cancer-related mortality worldwide. While drug treatments such as osimertinib are effective, they are rarely curative due to the eventual development of acquired resistance. Therefore, it is important to understand the mechanisms underlying this resistance and to identify strategies to overcome or prevent it. Ongoing studies at the San Raffaele Research Center showed that human lung cancer cells HCC827 display reduced level of Epidermal Growth Factor Receptor (EGFR) protein compared to the non-treated samples, suggesting a potential role for EGFR in drug-resistance development. In this study we aim to investigate the statistical significance of such decrease, its underlying causes, and the mechanisms by which cancer cells compensate for reduced EGFR activity. Analysis of RNA levels across six datasets of osimertinib-treatment response confirmed that EGFR down-modulation in post-treatment samples is not random but a consistent feature. Genomic investigation revealed a relative homozygous deletion in the post-treatment sample compared to the pre-treatment baseline, offering a genetic explanation for the protein reduction. While our work led to the hypothesis that *EGFR* gene deletion was caused by the down-modulation of translation-related genes, RNA analysis refuted this, showing no clear pattern of down-regulation. To understand survival mechanisms in the absence of EGFR, we developed a computational tool, which revealed the up-regulation of specific compensatory pathways. Furthermore, this tool identified key transcription factors involved in these pathways, highlighting potential new therapeutic targets to overcome resistance.

Lung cancer is a disease with an extremely high worldwide mortality rate, accounting for a significant proportion of new cancer diagnoses and deaths worldwide [1]. It is clinically classified into two main histological groups: small cell lung cancer (SCLC) and non-small cell lung cancer (NSCLC). Lung adenocarcinoma (LUAD) is the predominant subtype of NSCLC and the most common form of lung cancer globally [2].

Histologically, LUAD is characterized by glandular differentiation and mucus production in the peripheral regions of the lung. While smoking is a primary risk factor, LUAD is notably the most frequent histological subtype observed in never-smokers and former smokers [3].

At the molecular level, LUAD is a heterogeneous disease defined by recurrent "driver" mutations that confer a selective growth advantage and drive tumorigenesis. Among these, activating mutations in the *EGFR* gene are clinically pivotal, leading to the constitutive activation of downstream signaling cascades such as the MAPK and PI3K/AKT pathways [4]. The identification of these drivers has revolutionized treatment through the development of EGFR-targeted tyrosine kinase inhibitors (TKIs).

Despite the initial efficacy of EGFR-TKIs, acquired resistance inevitably emerges, driven by genomic and transcriptomic adaptations such as secondary mutations or compensatory signaling activation [5]. Consequently, transcriptomic profiling is critical for elucidating the mechanisms of EGFR down-modulation and resistance. Characterizing these expression changes is essential for identifying novel biomarkers and therapeutic strategies to improve long-term patient outcomes [6].

1.2. The EGFR Signaling Pathway in Normal and Cancer Cells

The EGFR is a transmembrane receptor tyrosine kinase of the ErbB family widely expressed in epithelial tissues where it regulates homeostasis [7]. Under physiological conditions, EGFR activation is strictly ligand-dependent. Binding of ligands such as EGF or TGF- α induces receptor dimerization and autophosphorylation of cytoplasmic tyrosine residues. The phosphorylated sites recruit adaptor proteins to initiate downstream cascades — primarily the RAS–RAF–MAPK pathway (driving proliferation) and the PI3K–AKT pathway (promoting survival) — ensuring controlled tissue repair and differentiation [8, 9].

In LUAD such regulatory control is frequently disrupted by somatic activating *EGFR* mutations, most notably exon 19 deletions and the L858R point mutation. These alterations result in constitutive, ligand-independent kinase activity, driving unchecked neoplastic growth [4]. This reliance on aberrant EGFR signaling, termed *oncogene addiction*, renders mutant tumors highly sensitive to TKIs. However, the efficacy of EGFR-TKIs is limited by the eventual emergence of acquired resistance. Tumor cells adapt through mechanisms such as the development of T790M secondary mutation, activation of bypass signaling tracks, or transcriptomic reprogramming [5]. Consequently, characterizing the genomic and transcriptomic landscape of EGFR signaling is critical for elucidating these adaptive survival mechanisms [10].

1.3. Genomic and Transcriptomic Approaches to Study Drug Resistance

To understand the evolution of drug resistance, it is necessary to interrogate molecular alterations at both DNA and RNA levels. Genomic approaches, utilizing high-throughput sequencing technologies, are the standard for detecting structural alterations and somatic mutations that confer resistance. In lung adenocarcinoma, such methods identified the EGFR T790M secondary mutation as the primary mechanism of resistance to early-generation TKIs [5, 11]. Complementarily, transcriptomic analyses (RNA-seq) capture the dynamic expression profiles of resistant cells. Unlike static genomic data, transcriptomics reveals functional adaptations, such as the up-regulation of compensatory kinases or the induction of epithelial-to-mesenchymal transition (EMT), which frequently drive resistance to third-generation inhibitors like osimertinib [12].

Integrating these multi-omics layers provides a comprehensive view of the tumor’s adaptive landscape, allowing researchers to distinguish between irreversible genetic loss and reversible transcriptional down-regulation of targets like EGFR. Such integra-

tive analyses are essential for uncovering complex resistance mechanisms and identifying actionable therapeutic targets that genomic data alone might overlook [13, 6].

1.4. Osimertinib: Mechanism and Clinical Application

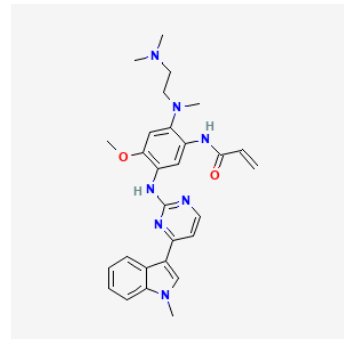


Figure 1: Chemical structure of osimertinib. PubChem Identifier: 71496458.

Osimertinib (Figure 1) is a third-generation, irreversible TKI designed to target NSCLC harboring activating *EGFR* mutations. Osimertinib forms a covalent bond with the C797 residue in the ATP-binding pocket of the EGFR kinase domain [14] inhibiting downstream oncogenic signaling cascades essential for cell proliferation and survival [15]. Clinically, osimertinib is distinguished by its ability to target both sensitizing mutations (e.g., L858R, Exon 19 deletions) and the T790M resistance mutation, which limits the efficacy of first- and second-generation inhibitors. Furthermore, its structural design confers high selectivity for mutant EGFR forms over wild-type, resulting in a more favorable toxicity profile [14].

2. Problem Definition and Solution Pipeline

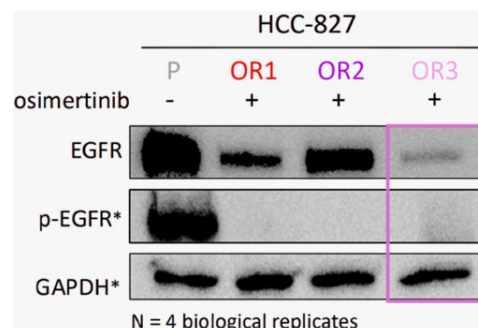


Figure 2: Western blot analysis (San Raffaele Institute) showing complete absence of EGFR protein in an osimertinib-resistant sample (highlighted), contrasting with expression in other resistant lines.

Preliminary western blot analysis provided by the San Raffaele Research Center (Milan) identified a distinct subset of osimertinib-resistant lung adenocarcinoma HCC827 cells characterized by the complete

absence of EGFR protein (Figure 2). Determining whether this phenotype is driven by irreversible genomic deletion or reversible transcriptional repression is critical to identify potential therapeutic biological targets.

2.1. Research Questions

The present study aims to investigate the molecular mechanisms underlying resistance to osimertinib in lung adenocarcinoma, with a specific focus on the down-modulation of EGFR. The research is guided by the following questions:

- What is the frequency of EGFR down-modulation in osimertinib-resistant lung adenocarcinoma, and to what extent does this phenomenon occur at the DNA and RNA levels? In particular, how often is loss of detectable EGFR protein observed following osimertinib treatment?
- Is EGFR down-modulation primarily associated with genomic alterations, such as gene loss or missense mutations, or with transcriptional changes leading to reduced EGFR mRNA expression?
- Which molecular mechanisms and regulatory factors are associated with EGFR down-modulation in the context of osimertinib resistance? Specifically, are there distinct gene expression signatures or recurrent EGFR mutations that correlate with this phenotype?

2.2. Solution Pipeline

To answer these questions, we developed a multi-omics computational pipeline (Figure 3).



Figure 3: Overview of the computational pipeline we proposed integrating data collection, differential gene expression analysis on transcriptomic data, copy number analysis on genomic data and compensatory pathway and transcription factor analyses.

3. Materials and Methods

3.1. Data Collection

The first step to address the research questions was the collection of publicly available genomic and transcriptomic datasets suitable for studying the resistance to

osimertinib in LUAD (Figure 3). Datasets were selected to include tumor samples or cell lines exposed to osimertinib and characterized by acquired drug resistance or persistence, along with untreated or sensitive controls.

3.1.1 Transcriptomic Datasets

A systematic search of cancer genomics repositories was conducted to identify studies on osimertinib resistance in LUAD. TCGA datasets were excluded due to the absence of osimertinib-treated cohorts; therefore, the analysis focused on six independent RNA-seq datasets retrieved from the Gene Expression Omnibus (GEO). The selected cohort comprises 4 studies utilizing HCC827 cell line models of acquired resistance and drug persistence (GSE278222, GSE202859, GSE249721, GSE283536), and 2 longitudinal datasets derived from patient biopsies collected at baseline and upon disease progression (Table A.3). Data retrieval was performed using the `GEOquery` R package. Gene expression matrices and sample metadata were manually curated to harmonize experimental definitions (e.g., Sensitive vs. Resistant) across studies, ensuring consistency for downstream comparative meta-analysis.

3.1.2 Genomic Data

Due to the lack of publicly available genomic datasets with matched pre- and post-treatment profiling, this study utilized proprietary sequencing data obtained through collaboration with the San Raffaele Scientific Institute. The dataset consists of genomic profiles from the HCC827 lung adenocarcinoma cell line, comprising both parental and osimertinib-resistant samples. These data were analyzed to identify structural alterations and mutations associated with the resistance phenotype, with a specific focus on characterizing the *EGFR* locus.

3.2. Transcriptomic Data Processing and Analysis

3.2.1 Pre-processing of Transcriptomic Data

To ensure comparability across independent studies, a harmonized pre-processing workflow was applied to all datasets. Initially, expression distributions were inspected to distinguish between raw counts and pre-normalized data. For datasets provided as raw counts, normalization was performed using the `DESeq2` median-of-ratios method to correct for differences in sequencing depth.

Subsequently, all expression matrices underwent $\log_2$ transformation to stabilize variance for statistical modeling. Quality control was rigorously maintained through the inspection of summary statistics

and distribution plots, confirming the efficacy of normalization prior to downstream differential expression analysis.

3.3. Differential Gene Expression Analysis

The next step (point 2 in Figure 3) was exploring changes in gene expression levels when cancer cells acquire osimertinib resistance to the *EGFR* gene. This was achieved through a systematic Differential Expression Analysis (DEA) applied to multiple independent datasets. Critically, each dataset was analyzed separately to accommodate its specific experimental design, ensuring that the results were statistically comparable for subsequent meta-analysis. Samples were rigorously categorized as "Sensitive" (untreated or drug-responsive) or "Resistant" (samples persisting after drug exposure).

The analytical framework selected depended on the data format:

- For **normalized expression data**, the `limma` package was used, employing linear models and empirical Bayes moderation.
- For **raw count data** (e.g., RNA-seq), the `DESeq2` framework was applied, which utilizes the negative binomial distribution and handles internal normalization.

Expression changes were quantified by the **log2 fold change** (log2 FC), and statistical significance was assigned. To control for false positives when testing thousands of genes, all *p*-values were subjected to the **Benjamini-Hochberg False Discovery Rate (FDR) correction**. This pipeline produces and adjusted *p*-values, which serve as the foundation for all subsequent analyses.

3.3.1 Meta-analysis of Differential Expression Results

To identify genes consistently associated with osimertinib resistance across the independent transcriptomic studies, a meta-analysis of differential expression results was performed. Given the heterogeneity of experimental designs, sample origins (cell lines versus human tissue), and sequencing protocols among datasets, a statistical aggregation approach based on study-level summary statistics was adopted.

Data Harmonization and Pre-processing For each dataset, DEA produced gene-wise statistics, including logFC and associated adjusted *p*-values. Prior to integration, datasets were harmonized by mapping all gene identifiers to standard Human Gene Nomenclature Committee (HGNC) gene symbols. Only genes detected in all six datasets were retained for the initial comparative analysis. Duplicate entries for the same gene within a single dataset were handled by retaining the entry with the most significant adjusted *p*-value.

Statistical Integration via Fisher's Method

The meta-analysis employed Fisher's combined probability test. For each common gene, *p*-values from the individual datasets were combined into a single test statistic (X^2), defined as:

$$X^2 = -2 \sum_{i=1}^k \ln(p_i)$$

where p_i is the adjusted *p*-value of the gene in the *i*-th dataset and k is the total number of datasets ($k = 6$). This method tests the null hypothesis that a gene is not differentially expressed in any of the studies against the alternative that it is differentially expressed in at least one.

Ranking and Functional Analysis To prioritize genes that were both statistically significant and directionally consistent, a custom ranking metric was calculated for each gene by multiplying the Fisher score by the sign of the average log2FC across all datasets. This ranked list was subjected to Gene Set Enrichment Analysis (GSEA) using MSigDB Hallmark and Gene Ontology (GO) Biological Process gene sets to identify consistently altered biological pathways.

Visualization and Network Analysis Results were visualized using a meta-volcano plot, plotting the average log2FC against the Fisher score. Heterogeneity of top candidates was assessed using forest plots and heatmaps. Finally, to investigate functional connectivity among the top-ranked genes, a Protein-Protein Interaction (PPI) network was constructed using the STRING database (v12.0) via its API, focusing on high-confidence interactions (score > 0.7) and *hub* gene identification.

3.4. Genomic Data Analysis

3.4.1 Genomic Data Processing

Consistent with the computational pipeline (Figure 3), raw paired-end sequencing data (FASTQ files) were processed using the standard GATK Best Practices workflow.

Reads were aligned to the human reference genome (GRCh38) using the **Burrows-Wheeler Aligner (BWA)**. The resulting alignments were converted to BAM format and sorted using **SAMtools**. To minimize technical artifacts, PCR duplicates were marked and removed using **Picard (MarkDuplicates)**. Finally, base quality score recalibration (BQSR) was performed using **GATK (BaseRecalibrator and ApplyBQSR)** to correct systematic errors in the quality scores assigned by the sequencer.

3.4.2 Copy Number Analysis

Allele-specific copy number alterations (CNAs) were inferred using FACETS (Fraction and Allele-Specific Copy Number Estimates from Tumor Sequencing). The analysis was conducted in matched mode, utilizing the post-treatment sample as the "tumor" and the pre-treatment sample as the "normal" reference baseline.

Aligned BAM files were processed to generate single-nucleotide polymorphism (SNP) pileups at germline heterozygous loci. These SNP counts served as input for FACETS to estimate tumor purity, ploidy, and allele-specific copy number profiles. This matched configuration enables the direct identification of acquired somatic events, including gains, deletions, and loss-of-heterozygosity (LOH).

Given the specific focus on *EGFR* down-modulation, downstream analysis was restricted to Chromosome 7. The integer copy numbers of major and minor alleles were extracted and visualized using Python (Matplotlib) to compare the genomic landscape of the post-treatment sample against the diploid baseline.

3.4.3 Analysis of Genes Strictly Related to *EGFR* Translation

To determine whether the observed *EGFR* down-modulation was driven by a global impairment of the translational machinery, we analyzed the expression levels of key genes governing protein synthesis.

Genes Selection Some genes contribute more to *EGFR* translation. Through literature research a set of genes was selected and the names are reported in Table A.4. The genes selected are related with the translation of genes with many 5' UTRs or are crucial for capping, involved in mTORC pathways and in pathways that at some point converge in mTORC: such characteristics are peculiar of *EGFR* gene. From the genes in the table, only the ones common to all the six datasets were considered for the analysis.

Plot of RNA Levels The RNA level of each gene was represented in an expression graph in order to see if *EGFR* reduction is caused by a down-modulation of these genes. The graph was implemented in Python using the libraries `pandas`, `numpy` and `matplotlib`.

3.5. Network Rewiring Analysis

Gene Expression Data and Differential Expression Analysis Finally, we implemented a tool for the discovery of *EGFR* compensatory pathways (Figure 3), which explain how cancer cells can survive without *EGFR*. Such tool also restitutes a

list of interesting transcription factors (TFs), that can be exploited as new drug target. The input of the tool is the RNA counts dataset GSE249721, imported into Python using `pandas`. For each gene, the mean expression was calculated for control samples (pre-treatment) and treated samples (post-treatment). Log2FC was defined like the difference between the mean of the pre-treatment and the post-treatment: $\log_2FC = \bar{E}_{\text{osi}} - \bar{E}_{\text{ctrl}}$. If $\log_2FC > 0$ genes were classified as up-regulated, if $\log_2FC < 0$ as down-regulated and if $\log_2FC = 0$ as neutral. To evaluate the significance of log2FC, the Welch t-test was used through `scipy.stats.ttest_ind` function. This test does not assume equal variance between the groups, so it is commonly used when there are less samples. Genes with missing expression values were excluded from the analysis and their p-value were set to missing. Since a large number of genes was considered, the p-values were corrected with false discovery rate (FDR) of Benjamini-Hochberg through `statsmodels.stats.multitest.multipletests`. Genes with $FDR < 0.05$ were considered statistically significant.

Pathway Enrichment Analysis On these differential expressed genes, two pathways enrichment were executed: one for up-regulated genes and one for down-regulated ones. The enrichments were performed in Python through `gseapy` package. The function used is `enrichr`, using the `KEGG_2021_Human` gene set collection and the parameter `organism` set to `Human`. Enriched pathways with an adjusted p-value < 0.05 were considered significant.

KEGG Pathways Functional Annotation The names of the pathways were exported in R. In R, the packages `KEGGREST` and `string` were installed through `BiocManager` and used to query KEGG and do strings name manipulation. In fact, the pathway names were standardized by transforming them in lowercase, removing the text between parenthesis, eliminating the species suffix and changing non-alphanumeric characters. Each name was processed using `KEGGREST:keggFind`, prioritizing searches restricted to human pathways and expanding to broader searches when no matches were found. When there were more options, the best match was selected based on Levenshtein distance between the normalized query and the different name options. The retrieved name was query with `KEGGREST:keggGet` to achieve metadata and obtain the field `KEGG CLASS` in order to get the function of the pathways. If there were no results, the value "no available function" was assigned. The result is a file with the name of the pathways and their respective function. Such R object was imported in Python through `pandas`.

EGFR-Associated and Compensatory Pathway Analysis The next step was the analysis of the pathways, in particular filtering to see if EGFR was in up or down-regulated pathways. Then the compensatory pathways were identified. To visualize the function distribution of the EGFR down-modulated and compensatory pathways, two bar plots were generated thanks to `matplotlib`. The tool was implemented to handle also the opposite situation, when EGFR is up-regulated.

Transcription Factors Enrichment Analysis To identify TFs that could be involved in pathways alterations, two enrichments were executed. The first TFs enrichment analysis was performed in Python with the function `enrichr` from `gseapy` package on differential expressed genes; the second enrichment was executed with `VIPER` in R. To implement the first enrichment, more TFs libraries were exploited; these libraries were searched through keyword matching. The selected libraries included ChEA, ENCODE TF ChIP-seq, TRANSFAC, JASPAR, TRRUST e ARCHS4 TF co-expression. Only TFs with an adjusted p-value < 0.05 were retained and duplicates were removed. The second enrichment was executed with `VIPER` to retrieve also the TFs activity. `VIPER` requires two inputs: the expression matrix with genes as rows, samples as columns; the second input is the regulon: TF-target interactions obtained from DoRoThEA for humans, maintaining only interactions with confidence levels A, B and C. The TFs activity was measured using Normalized Enrichment Scores (NES) and the activity depends on coordinate expression of targets. Then TFs were filtered for p-value < 0.05 and $|\text{NES}| > 2.0$. Finally, the TFs of the two enrichments were merged, and only the common TFs were conserved. From these retained TFs, two classes were defined:

- **Class A:** TFs showing both a significant change in activity and differential expression.
- **Class B:** TFs showing a significant change in activity but not differentially expressed.

The two classes were graphically visualized through lollipop plots realized with `matplotlib`.

4. Results

4.1. Meta-analysis of Transcriptomic Datasets

Differential expression results from six independent RNA-seq studies were integrated to identify consistent gene expression changes associated with osimertinib resistance.

4.1.1 Gene Overlap Across Studies

Filtering for genes present and tested in all six datasets yielded a common set of **4,638 genes** for meta-analysis. Figure 4 illustrates the gene overlap across the independent datasets.

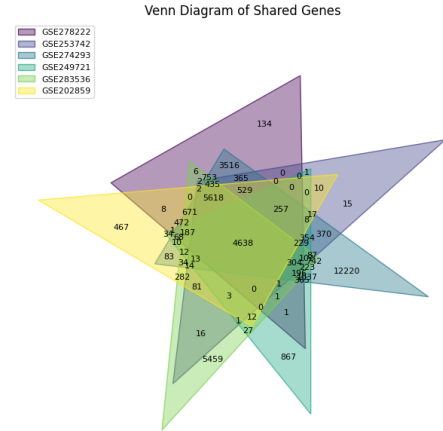


Figure 4: Overlap of genes detected across the six transcriptomic datasets. The intersection represents the 4,638 genes shared across all studies.

4.1.2 Aggregation of Statistical Evidence

Adjusted p-values for differential expression were aggregated across the six datasets for the 4,638 shared genes. Figure 5 displays the distribution of these aggregated significance scores, showing a stratification between genes with weak evidence and a subset exhibiting consistent statistical signals across studies.

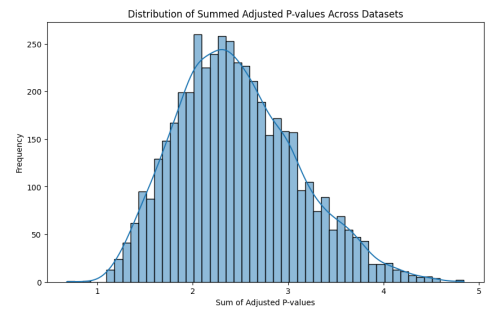


Figure 5: Distribution of aggregated adjusted p-values across the 4,638 genes shared among all datasets.

4.1.3 Consistency of EGFR Differential Expression

We evaluated EGFR differential expression across the six individual studies (Figure 6). A consistent trend toward reduced expression in resistant or persistent samples compared to sensitive controls was observed, though effect sizes and significance levels varied across datasets.

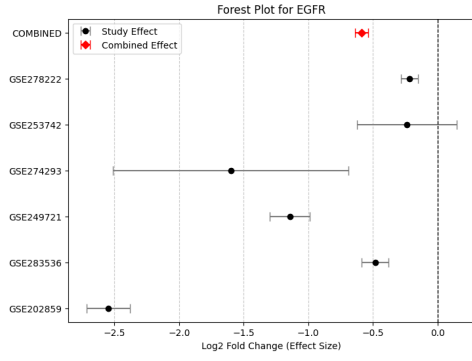


Figure 6: Forest plot of EGFR differential expression across the six transcriptomic datasets. Points represent estimated effect sizes with corresponding confidence intervals.

4.1.4 Meta-analysis of Differential Expression

Genes were ranked by aggregated significance after multiple testing correction. Figure 7 displays expression patterns for selected genes, showing consistent EGFR down-regulation IN CHE CAMPIONI across studies.

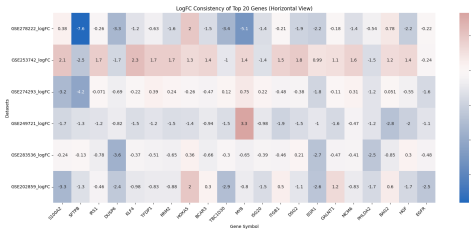


Figure 7: Meta-analysis heatmap of differential expression patterns for selected genes, highlighting EGFR.

Global meta-analysis results are summarized in Figure 8. EGFR is highlighted among genes exhibiting both strong statistical support and large expression changes.

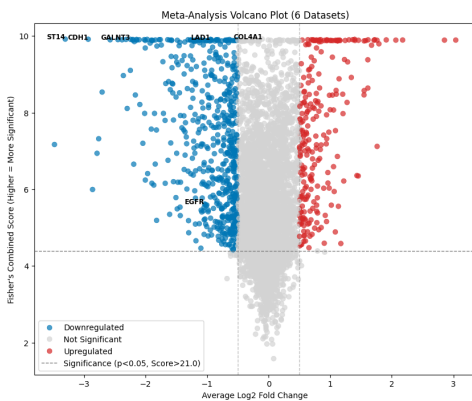


Figure 8: Meta-analysis volcano plot of aggregated differential expression results. EGFR is indicated among genes with significant transcriptional changes.

4.2. Pathway Enrichment and Network Analysis

4.2.1 Gene Set Enrichment Analysis

Gene Set Enrichment Analysis (GSEA) was applied to the ranked list of differentially expressed genes. Figure 9 and Table A.1 summarize the top enriched pathways.

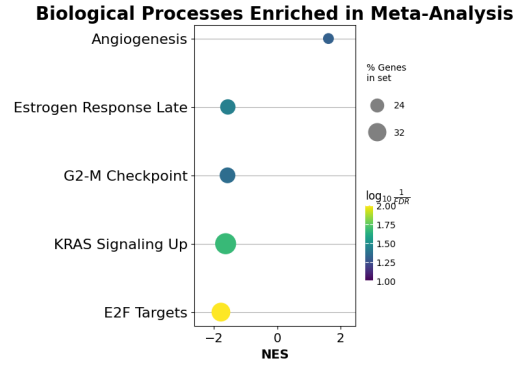


Figure 9: Gene Set Enrichment Analysis (GSEA) results based on meta-analysis differential expression ranking. The top enriched pathways are shown with their normalized enrichment scores (NES) and FDR-adjusted q-values.

Pathways showing negative normalized enrichment scores (NES) included *E2F Targets* (NES = -1.77, FDR $q = 0.005$), *G2-M Checkpoint* (NES = -1.57, FDR $q = 0.044$), *KRAS Signaling Up* (NES = -1.62, FDR $q = 0.021$), and *mTORC1 Signaling*. Positive enrichment was observed for pathways such as *Angiogenesis* (NES = 1.62, FDR $q = 0.049$) and *Epithelial-Mesenchymal Transition*.

4.2.2 Protein-Protein Interaction Network

From STRING we obtained the protein-protein interaction network shown in Figure 10. Table A.2 lists the top ten hub genes. *MYC* was the most connected node (239 connections), followed by *EGFR* (199 connections). Additional hubs included cell cycle regulators (*CDK1*, *BRCA1*), stress response proteins (*HSP90AA1*, *HSPA4*), and signaling mediators (*ATM*, *CDH1*, *CD44*).

4.3. Copy number Alterations Associated with Osimertinib Resistance

Figure 11 illustrates the allele-specific copy number (CN) of the *EGFR* locus in the post-treatment sample, normalized against the pre-treatment baseline.

Since this analysis is relative to the pre-treatment state, the results isolate acquired somatic events, specifically highlighting a homozygous deletion that emerged during treatment. It is important to note that these values reflect changes relative to the baseline rather than raw absolute counts. To ensure biological accuracy, the model was corrected for tumor

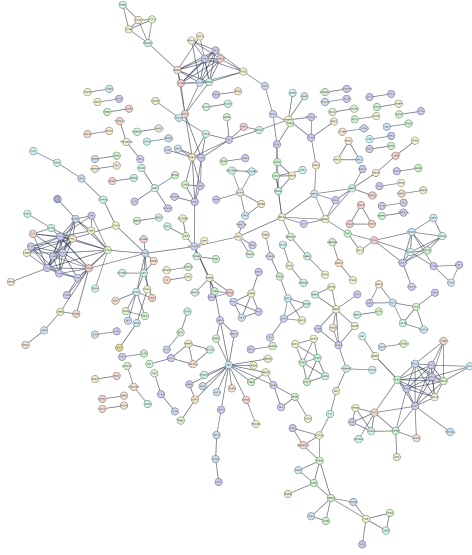


Figure 10: Protein-protein interaction network generated using STRING for genes associated with osimertinib resistance. Nodes represent proteins, and edges represent high-confidence interactions. Highly connected hub proteins are highlighted.

purity ($\sim 44\%$) and ploidy (3.8), as estimated by the FACETS algorithm.

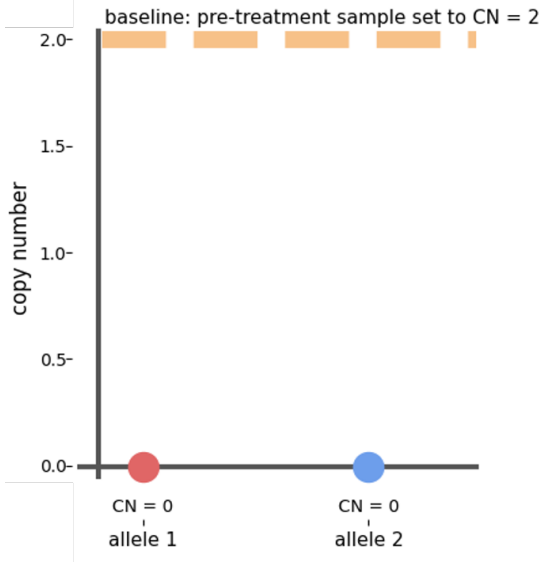


Figure 11: The graph shows the relative copy number of each allele in post treatment sample. There are two fewer copies.

4.4. Expression of EGFR Translation and Stability Regulators

To evaluate the potential contribution of translational defects to the observed phenotype, the expression profiles of genes governing EGFR translation and stability were analyzed (Figure 12).

The data do not exhibit a distinct pattern of down-regulation across the six independent datasets. Most regulatory factors maintained stable expression levels in resistant samples compared to controls. This observation suggests that the reduction of EGFR protein is not driven by a transcriptional decline in the canonical

translational machinery.

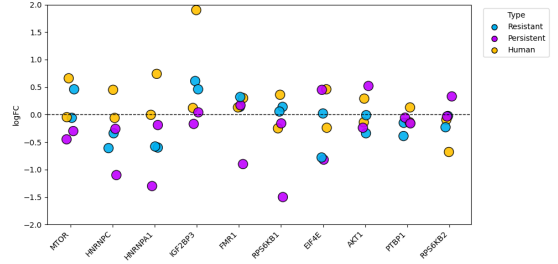


Figure 12: Heatmap displaying the differential expression of genes regulating EGFR translation and stability.

4.5. Compensatory Signaling Pathways Activated in Resistant Samples

Figure A.1 displays the functions performed by EGFR down-modulated pathways, with ten pathways linked to signal transduction function. Figure 13 shows that signal transduction is indeed the most compensated function through the four pathways reported in Table 1. TGF-beta signaling is the most important pathway identified, as it is already associated with EGFR-TKI resistance and invasive capacity [16].

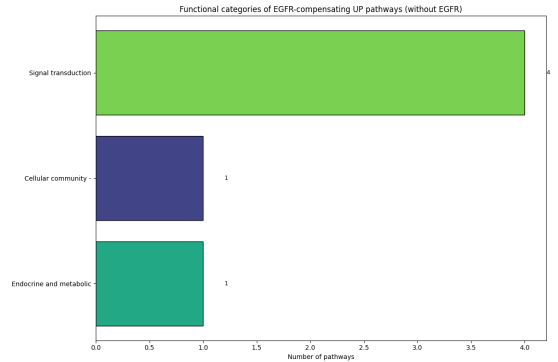


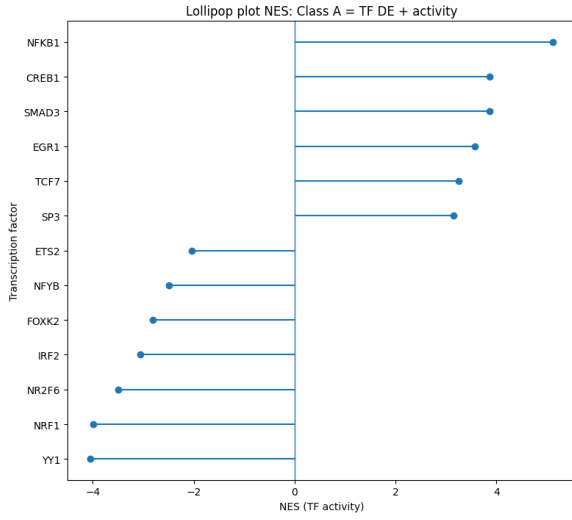
Figure 13: The bargraph shows the compensated functions and the respective number of EGFR independent and up-modulated pathways associated with.

Table 1: Significantly enriched signaling compensatory pathways associated with EGFR down-modulation. For each pathway, the overlap between the input gene list and the pathway gene set, the adjusted p-value, and the assigned KEGG functional category are reported.

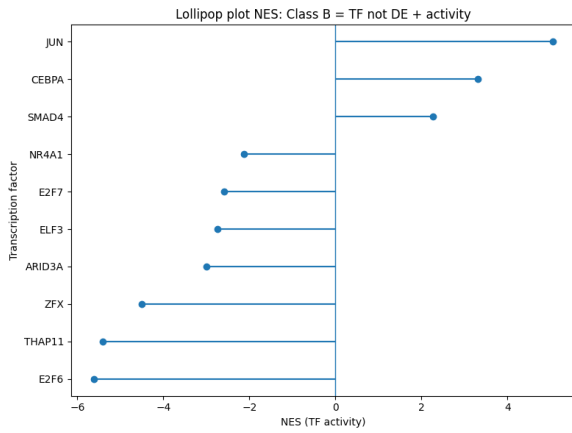
Term	Overlap	Adjusted p-value	Function
TGF-beta signaling pathway	13/94	0.017302	Signal transduction
Ras signaling pathway	21/132	0.028553	Signal transduction
MAPK signaling pathway	24/294	0.028553	Signal transduction
Rap1 signaling pathway	19/210	0.028553	Signal transduction

4.6. Transcription Factors Associated with EGFR Down-modulation

Figure 14 shows two lollipop plots, one for class A and one for class B TFs. Factors in class B may be more valid for further analysis, considering that their change in activity depends on something happened upstream.



(a) Transcription factors showing both significant differential expression and activity changes (Class A).



(b) Transcription factors showing significant activity changes without differential expression (Class B).

Figure 14: VIPER-inferred transcription factor activity in treated versus control samples. NES values indicate increased or decreased TF activity.

5. Discussion

This study integrates transcriptomic and genomic analyses to investigate mechanisms of osimertinib resistance in lung adenocarcinoma.

A meta-analysis across six independent RNA-seq datasets revealed consistent down-regulation of EGFR mRNA following osimertinib treatment. As illustrated in Figure 6 and Figure 8, EGFR is characterized by a robust decrease coupled with strong statistical significance, distinguishing it as a recurrent molecular feature of the resistant phenotype. These findings indicate that EGFR transcriptional loss represents a recurrent feature of osimertinib resistance rather than a dataset-specific effect. In addition, pathway-level analyses further identified enrichment of biological processes 9 previously linked to drug resistance like EMT [17] and KRAS signaling [18], suggesting that EGFR down-modulation happens in a systematic way. The **STRING** output (Figure 10) shows that EGFR is a hub gene, together with key regulators like MYC and

HSP90AA1, highlighting EGFR involvement in coordinating downstream signaling responses.

Genomic copy number analysis suggested a relative reduction in EGFR copy number following treatment; however, limited sample size and low tumor purity prevent definitive conclusions regarding its contribution to EGFR protein loss. This reduction is not caused by a down-modulation of genes strictly related with EGFR translation, as shown in Figure 12, where no evident pattern of down-regulation can be appreciated. The genes of this graph are the genes common to all the six datasets and we selected them because they influence EGFR translation. We chose core components of the cap-dependent translation machinery which include EIF4E [19]; then we selected MTOR, AKT1, RPS6KB1/2 because they are key regulators of the mTORC1 pathway and this pathways control EIF4F and its component [20]; and then we took RNA-binding proteins implicated in translation control like IGF2BP3 [21], HNRNPA1 [22], PTBP1 [23], and FMR1 [24].

Network and transcription factor analyses further indicate that resistance is associated with signaling rewiring and activation of EGFR-independent regulatory programs (Figure 13, highlighting the complexity of resistance mechanisms beyond single-gene alterations. The most interesting result between the compensatory pathways is TGF-beta signaling (Table 1); in fact, some pathways, including TGF-beta signaling and MAPK signaling (another pathway resulting from the tool), contribute to EMT, which allows cells to improve their resistance to treatment and their ability of invasion [16]. TGF-beta signaling pathway is also one result of the meta-analysis, confirming that the tool gives reliable results. Furthermore, the tool provides two lists of interesting TFs, one list contains TFs of class A and the other the TF of the class B (Figure 14. The change in activity of class A TFs (Figure 14a can be explained by the fact that they are differential expressed, instead class B TFs (Figure 14b) change in activity is more interesting because it is probably caused by something happened upstream. In particular in Figure 14b is present JUN, a transcription factor associated with gefitinib (a second generation LUAD drug) resistance in literature [25], and that is crucial for cells proliferation in other types of tumors [26].

6. Conclusion

In conclusion, this study provides evidence that EGFR down-modulation at the transcriptomic level is a consistent hallmark of osimertinib resistance in lung adenocarcinoma. Meta-analysis across heterogeneous datasets strengthens this observation and demonstrates the utility of integrative transcriptomic ap-

proaches to identify robust resistance-associated signatures. All the results from the meta-analyses lead us to assume that down-regulation is not a random event, but a crucial step in EGFR-TKI resistance.

While genomic alterations may contribute to EGFR loss, the available data suggest that resistance arises from a combination of transcriptional down-regulation and compensatory signaling mechanisms. These findings support a model in which osimertinib resistance is driven by network-level adaptation rather than a single dominant molecular event.

7. Limitations and Future Perspectives

Future work should incorporate additional transcriptomic and genomic datasets to increase statistical power and enable more precise estimation of EGFR copy number changes. Integration of quantitative proteomic data would be critical to determine whether EGFR loss occurs at the transcriptional, translational, or post-translational level.

Further refinement of pathway and TFs analyses, including integration with functional dependency resources such as DepMap or AI automated literature mining approaches, may improve the identification of actionable resistance mechanisms. These efforts could support the development of alternative therapeutic strategies targeting EGFR-independent biological factors in osimertinib-resistant lung adenocarcinoma.

8. Author contributions

Table 2: Contribution overview by task

Task Description	Contributors
Dataset Research & Pre-processing	Alvise Celeste Cova
Reference Dataset Table Construction	Alvise Celeste Cova
EGFR Frequency Pipeline Analysis Design	Alvise Celeste Cova, Nicolò Sevegnani
EGFR Frequency Analysis Pipeline Implementation	Alvise Celeste Cova
DNA Analysis Pipeline – Design	Muserref Yuca
DNA Analysis Pipeline – Implementation	Muserref Yuca
Translation Analysis Pipeline – Design	Nicolò Sevegnani
Translation Analysis – Implementation	Muserref Yuca, Alvise Celeste Cova
Tool Concept & Methodology Design	Nicolò Sevegnani
Tool Development	Nicolò Sevegnani
Slide Design & Assembly	Nicolò Sevegnani

A. Appendix

Table A.1: Top enriched pathways identified by GSEA. Pathways are ranked by statistical significance.

Pathway	NES	FDR q-value
E2F Targets	−1.77	0.005
KRAS Signaling Up	−1.62	0.021
Estrogen Response Late	−1.56	0.037
G2–M Checkpoint	−1.57	0.044
Angiogenesis	1.62	0.049
Fatty Acid Metabolism	1.51	0.056
Epithelial–Mesenchymal Trs	1.55	0.060
Wnt–β-Catenin Signaling	1.65	0.068
Notch Signaling	1.51	0.069
mTORC1 Signaling	−1.46	0.105

Table A.2: Top ten hub genes identified in the protein–protein interaction network.

Gene	Number of Connections
MYC	239
EGFR	199
HSP90AA1	153
BRCA1	139
UBC	132
CDK1	131
HSPA4	125
CDH1	115
CD44	104
ATM	101

Table A.3: Summary of transcriptomic datasets included in the meta-analysis.

Accession ID	Sample Type	Source	Resistance Phenotype
<i>Cell Line Models</i>			
GSE278222	Cell Line	HCC827	Drug Resistance
GSE202859	Cell Line	HCC827	Drug Resistance
GSE249721	Cell Line	HCC827	Drug Persistence
GSE283536	Cell Line	HCC827	Drug Persistence
<i>Patient Cohorts</i>			
GSEXXXXX	Patient Tissue	LUAD Biopsy	Acquired Resistance
GSEYYYYY	Patient Tissue	LUAD Biopsy	Acquired Resistance

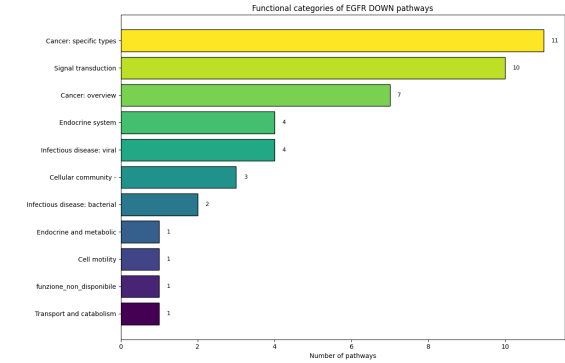


Figure A.1: This bar graph shows the inhibited function and the respective number of EGFR down-modulated pathways associated with.

Table A.4: Selected genes which influence EGFR translation

Selected genes			
EIF4E	MTOR	IGF2BP3	4EBP1
EIF4A	RPS6KB1	HNRNPA1	AKT1
EIF4G	RPS6KB2	HNRNPC	mTORC1
LARP1	PTBP1	FMR1	IGF2BP1

List of Abbreviations

CN = Copy Number
DEA = Differential Expression Analysis
FDR = False Discovery Rate
GEO = Gene Expression Omnibus
GO = Gene Ontology
EGFR = Epidermal Growth Factor Receptor
EMT = Epithelial-Mesenchymal Transition
LOH = Loss of Heterozygosity
LUAD = Lung Adenocarcinoma
KEGG = Kyoto Encyclopedia of Genes and Genomes
NES = Normalized Enrichment Score
NSCLC = Non-Small Cell Lung Cancer
SCLC = Small Cell Lung Cancer
TF = Transcription Factor
TCGA = The Cancer Genome Atlas
TKI = Tyrosine Kinase Inhibitor
HGNC = HUGO Gene Nomenclature Committee
FACETS = Fraction and Allele-Specific Copy Number
Estimates from Tumor Sequencing
BAM = Binary Alignment Map
BWA = Burrows-Wheeler Aligner
CNA = Copy Number Alteration
GATK = Genome Analysis Toolkit
GSEA = Gene Set Enrichment Analysis
logFC = log 2 Fold Change
MSigDB = Molecular Signatures Database
PPI = Protein-Protein Interaction
RNA-seq = RNA Sequencing
SNP = Single Nucleotide Polymorphism
UTRs = Un Translated Regions
TKI = Tyrosine Kinase Inhibitor
BQSR = Base Quality Score Recalibration
GRCh38 = Genome Reference Consortium Human
Build 38
MYC = Myelocytomatosis Viral Oncogene Homolog
KRAS = Kirsten Rat Sarcoma Viral Oncogene
Homolog

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